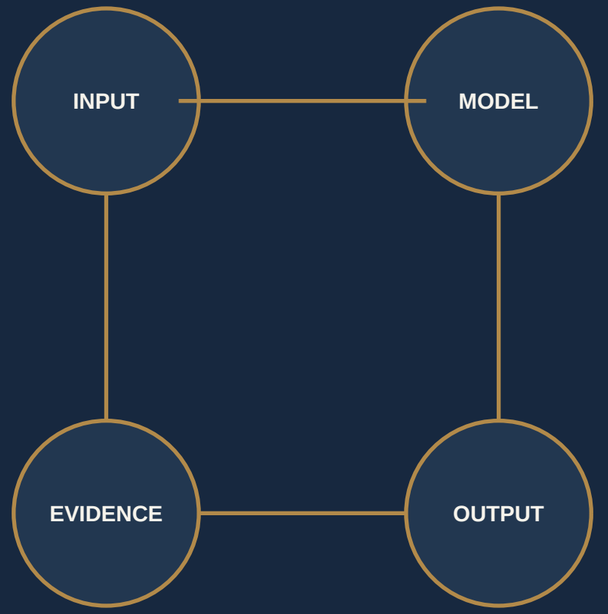
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Robot Navigation

AI.407 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define obstacle avoidance.
- Apply key methods including localization.
- Build or analyze a practical system involving mapping.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Obstacle avoidance

02

METHODS

Localization

03

SYSTEMS

Mapping

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Obstacle avoidance

Obstacle avoidance

02 · Localization

Localization

03 · Mapping

Mapping

04 · Global navigation

Global navigation

05 · Local navigation

Local navigation

06 · Reactive navigation and auton

Reactive navigation and autonomous navigation architectures

07 · Mandatory practical: Program a

Mandatory practical: Program a simulated autonomous robot to navigate unknown and dynamic environments

Core ideas, made explicit

AI.407 • Distinguish definitions, assumptions and evidence

- Obstacle avoidance
- Localization
- Mapping
- Global navigation
- Local navigation

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.407 • Distinguish definitions, assumptions and evidence

- Reactive navigation and autonomous navigation architectures
- Mandatory practical: Program a simulated autonomous robot to navigate unknown and dynamic environments

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Obstacle avoidance

INPUTS
Localization

METHOD
Mapping

OUTPUT
Global navigation

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving obstacle avoidance.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using obstacle avoidance.
- Implement or simulate a baseline using localization.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.